Austin Hutchen

I'm a Computer Science student at the University of Colorado Boulder who specializes in building cross-platform software, e-commerce websites, SEO, digital marketing, and crafting embedded scientific instruments.

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austinmorozane

Education

BS **University Of Colorado Boulder**, Computer Science — Boulder, CO, Aug 2023 – May 2028

- GPA: 3.097
- Built several projects including a quartz-powered oscillator clock, LED animation wall, automatic plant watering system, user vibe-based location search API, calculator app, note app, and more. Check out github and my website for the full list of (35) projects.

BS **University Of Colorado Boulder**, Mathematics — Boulder, CO, Aug 2026 – May 2028

- GPA: 3.17
- Recieved letter of recommendation for honors working as Calculus 2 Course Assistant.

MN **University Of Colorado Boulder**, Computer Engineering — Aug 2026 – May 2028

Experience

FormFactor, Software Engineering Intern — Gunbarrel, CO, May 2024 – Aug 2024, 4 months

Working as an embedded software engineer, collaborated with engineering team and wrote tests for a firmware simulator built on the MODBUS TCP standard.

- Developed a comprehensive ModBus ASCII/RTU library in C#, including simulators and extensive tests.
- Designed and tested a NUGET driver for interfacing with an RS-485 TCP server and an embedded temperature controller instrument (MOXA CND3) used in a quantum computer for near-zero Kelvin operations.
- Collaborated directly with the FormFactor Cryo Boulder team on the IQ-3000 Spin-Qubit Quantum Computer system.
- Using documentation, optimized ASCII/RTU algorithm for efficient retrieval of data on slower networks over TCP.

University Of Colorado Boulder, Course Assistant — Boulder, CO, Aug 2023 – Dec 2023, 5 months

Taught alongside faculty and held office hours weekly throughout the semester, to assist students struggling in APPLIED and MATH 2400 at CU Boulder.

- Developed visualization strategies for solving complex integrals, such as u-sub, volumes of solids, series, and inverse functions.
- Assisted with in-class recitations, in which students solved worksheets of related chapter problems weekly.
- Went to weekly meetings with the MATH department to discuss approaches for helping students with common problems in the course.

Mobius Materials, Digital Marketing Analyst | San Francisco, CA | May 2021 – Aug 2021 | 4 months

Worked as leader of marketing team, viewing weekly analytics and using them to further the development of online marketplace and quality of marketing material.

- Developed an article about the importance of regulating and developing tools to reuse e-waste Why Sustainability Ultimately Begins With E-Waste, Not Paper Straws. which led to the company's first major sale of repurposed components.
- Assisted Software Engineering team with development of e-waste marketplace UI.
- Led e-mail marketing campaigns, and wrote several drafts articulating the importance of a circular economy to interested buyers.

Projects

TheVibeCheck @ | Sept 2024 – Nov 2024

Open-source application for returning nearby locations based on the user's inputted mood. Working on a team, I integrated the google maps Places API, managed the CSS, and developed the main scripts.

- Achieved 2.8x speedup over baseline application launch on slower wifi connections by helping to optimize the SQL backend and CSS.
- Adopted by 3 contributors
- Demo viewable on website, on display at link

WebCam @ | June 2024 – Sept 2024

A webcam application similar to photobooth that captures realtime video, and compresses it using the BLOB format. Allows for upload to an Amazon FireBase database for storage and retrieval post-compression.

- Reduced streaming lag and file size by 50% with less than 10% accuracy degradation from initial release

Ecommerce Site and App @ | Feb 2024 – May 2024

A web application I built on request for a client, based on a wordPress original. Has built-in site transitions, support for mailChimp, and add to cart, purchasing functionality built-in.

- Added optimization helpers for functions like hamburger menu, main site pager animations, and the loading of large image arrays.
- Reduced bundle size by 20% by using a main scroll image loader, and optimized helpers for embedded site operations like purchasing and viewing object descriptions.

Calculator App @ | July 2021 – June 2023

An IOS calculator app built from scratch using C and Flutter, that supports transcendental functions. The UI was designed to look familar, similar to the default IOS calculator app.

- Added optimization helpers for functions like sqrt(), sin(), arctan(), pi, and similar operations.
- Reduced bundle size by 30% by using a main switch function, and optimized helpers for more complex operations.

Skills

Languages: C#, TypeScript, JavaScript, CSS, Java, SQL, Python, C++, CUDA, Rust, Julia

Infrastructure: Kubernetes, Docker, FireBase, AWS

Research Areas: Embedded Software Engineering, Firmware Development, Software Development, Computer Systems, Mathematics, and Mobile Development

Invited Talks

1. Group Theory, Abstract Algebra, and teaching complex Mathematics The Quaternionic Learning Process